

Technology Stack

Date	30 September 2026
Team ID	XXXXXX
Project Name	Learnmate – AI StudyBuddy
Maximum Marks	2 Marks

Define Your Technology Stack:

The programming languages, frameworks, databases, back-end tools, and APIs used to build Learnmate are listed below. A well-chosen technology stack ensures that the application is scalable, maintainable, and aligned with the project requirements.

Each row gives the technology chosen for a component of the architecture and a brief justification for the choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js (client served at <code>http://localhost:3000</code>)	Component-based UI for sign-up, login, file upload and study screens (summary, flashcards, quiz, study plan). The API allows this origin through CORS (CLIENT_URL).
2	Backend / Server-Side	Node.js with Express.js (express-async-errors, cors, dotenv)	Fast, non-blocking JavaScript runtime with a lightweight REST framework. Routes are split into <code>/api/auth</code> , <code>/api/materials</code> and <code>/api/admin</code> ; async errors reach one global error handler.
3	Database / Data Storage	MongoDB with Mongoose	Flexible document model suits nested study data (flashcards, quiz options) stored inside each material. Mongoose provides schemas, validation, timestamps and unique email checks.
4	Cloud / Hosting / Deployment	Local development server (Node.js on port 5000); MongoDB connection via MONGO_URI	Configuration through environment variables keeps the app portable, so it can be deployed to any Node.js host with a MongoDB Atlas database.
5	Version Control & CI/CD	Git and GitHub	Source control and team collaboration through repositories, commits and pull requests. No automated CI/CD pipeline is set up yet.
6	Third-Party APIs / Other Tools	Google Gemini API (@google/generative-ai, model gemini-3.7-flash)	Generates the summary, flashcards, quiz and study plan from uploaded text. Calls are retried up to 3 times with exponential backoff on 429/503 errors.
7	Authentication & Security	JSON Web Tokens (jsonwebtoken) and bcryptjs	Stateless login with signed access tokens, role-based access (student / admin) and password hashing (10 salt rounds) before saving to the database.
8	File Upload Handling	Multer (disk storage)	Accepts .txt, .md and .pdf files up to 5 MB, validates the file type, and saves each file with a unique name in the uploads folder.

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9	API Testing	Postman	Used to test and document the authentication, materials and admin endpoints during development.